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DISPLAY PANEL

Abstract

A display panel includes a substrate and a plurality of pixels on the substrate. Each of the plurality of pixels includes a plurality of sub-pixels coaxially stacked. Each of the plurality of sub-pixels includes a conductive layer, a light-emitting layer connected to the conductive layer, and a base layer. The plurality of pixels arranged in an array on a plane vertical to a stacking direction of the plurality of sub-pixels. The conductive layer is configured to transmit electrical signal. The light-emitting layer is configured to emit light according to the electrical signal. The base layers of the sub-pixels with a same sequence along the stacking direction are connected with each other.

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Background/Summary

FIELD

[0001] The subject matter herein generally relates to display apparatus, and particularly relates to a display panel.

BACKGROUND

[0002] A conventional display panel includes multiple pixels emitting light together to display images. Each of the pixels is composed by multiple sub-pixels, and each of the sub-pixels may be a red light-emitting diode (LED), a green LED, or a blue LED. For display panels of a same size, the more the pixels, the better an image quality and a color quality. Due to a coplanar arrangement of the pixels, it is difficult to reduce the size of the display panel, and manufacturing a small display panel based on the coplanar arrangement have a complex process and a low yield.

[0003] Therefore, there is room for improvement within the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Implementations of the present disclosure will now be described, by way of embodiment, with reference to the attached figures, wherein:

[0005] FIG. 1 is a schematic view of a planar structure of a display panel according to an embodiment of the present application.

[0006] FIG. 2 is a partial cross-sectional view of a display panel according to a first embodiment of the present application.

[0007] FIG. 3 is a partial cross-sectional view of a display panel according to a second embodiment of the present application.

[0008] FIG. 4 shows waveforms of pulse modulation signals from one driving circuit for driving sub-pixels of the display panel according to a first embodiment.

[0009] FIG. 5 shows waveforms of pulse modulation signals from multiple driving circuits for driving sub-pixels of the display panel according to a first embodiment.

DETAILED DESCRIPTION

[0010] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts have been exaggerated to better illustrate details and features of the present disclosure.

[0011] Several definitions that apply throughout this disclosure will now be presented.

[0012] The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like.

[0013] “Above” means one layer is on top of another layer. In one example, it means one layer is situated directly on top of another layer. In another example, it means one layer is situated over the second layer directly or indirectly with more layers or spacers in between.

[0014] When a feature or element is herein referred to as being “on” another feature or element, it

can be directly on the other feature or element or intervening features and/or elements may also be present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached, or coupled to the other feature or element or an intervening features or elements may be present.

First Embodiment

[0015] FIG. 1 shows a display panel **100** of the first embodiment includes a substrate **10** and a plurality of pixels **20** on a same surface of the substrate **10**. The substrate **10** includes a driving substrate **11**.

[0016] Each of the plurality of pixels **20** includes a plurality of sub-pixels **30**. The plurality of sub-pixels **30** are coaxially stacked in a stacking direction parallel to a thickness direction of the plurality of sub-pixels **30**. Sub-pixels **30** in a same pixel **20** are used to emit image light of different colors according to a time sequence, thereby enabling the display panel **100** to display color images. The plurality of pixels **20** are arranged in an array perpendicular to the stacking direction of the plurality of sub-pixels **30**, and a size of each pixel **20** is between 1 μm ~500 μm .

[0017] Referring to FIG. 2, each sub-pixel **30** includes a conductive layer **31**, a light-emitting layer **32**, and a base layer **33**. The conductive layer **31** is used to transmit electrical signals. The light-emitting layer **32** is connected to the conductive layer **31** for emitting light. The base layer **33** is fixed on a side of the light-emitting layer **32** away from the conductive layer **31**.

[0018] Along the stacking direction, the base layers **33** of the sub-pixels **30** with a same sequence are connected with each other. For example, the base layers **33** of the sub-pixels **30** at the top are connected to each other to form an entire layer, the base layers **33** of the sub-pixels **30** at the middle are connected to each other to form an entire layer, and the base layers **33** of the sub-pixels **30** at the bottom are connected to each other to form an entire layer. The sequence of each sub-pixel **30** indicates an order number along the stacking direction of the corresponding sub-pixel **30** in the pixel **20**. The stacking direction can be from top to bottom or from bottom to top.

[0019] In this embodiment, each pixel **20** of the display panel **100** includes three sub-pixels **30**, namely the first sub-pixel **301**, the second sub-pixel **302**, and the third sub-pixel **303**, which are sequentially coaxial stacked for emitting a first light, a second light, and a third light.

[0020] Base layers **33** of all the first sub-pixels **301** are connected with each other to form an entire layer. That is, the conductive layers **31** and the light-emitting layers **32** of all the first sub-pixels **301** are on a same layer. Similarly, base layers **33** of all the second sub-pixels **302** are connected with each other to form an entire layer, and base layers **33** of all the third sub-pixels **303** are connected with each other to form an entire layer. In each pixel **20**, the first sub-pixel **301** is on the driving substrate **11** of the substrate **10**, the conductive layer **31** and the light-emitting layer **32** of the first sub-pixel **301** are between the base layer **33** and the driving substrate **11** of the first sub-pixel **301**, the second sub-pixel **302** is on a side of the base layer **33** of the first sub-pixel **301** away from the driving substrate **11**, the conductive layer **31** and the light-emitting layer **32** of the second sub-pixel **302** are between the base layer **33** of the second sub-pixel **302** and the base layer **33** of the first sub-pixel **301**, the third sub-pixel **303** is on a side of the base layer **33** of the second sub-pixel **302** away from the first sub-pixel **301**, and the conductive layer **31** and the light-emitting layer **32** of the third sub-pixel **303** are between the base layer **33** of the third sub-pixel **303** and the base layer **33** of the second sub-pixel **302**.

[0021] In this embodiment, each sub-pixel **30** is a miniature inorganic light-emitting diode, and the light-emitting layer **32** is an inorganic light-emitting layer. In this embodiment, the light-emitting layer **32** includes a buffer layer **321**, an N-type gallium nitride (GaN) layer **322**, a quantum well light-emitting layer **323**, a P-type GaN layer **324**, a transparent conductive layer **325**, and a distributed Bragg reflector (DBR) layer **326**. The buffer layer **321** is on a side of the base layer **33**, the N-type GaN layer **322** is on a side of the buffer layer **321** away from the base layer **33**, the quantum well light-emitting layer **323** is on a side of the N-type GaN layer **322** away from the

buffer layer **321**, the P-type GaN layer **324** is on a side of the quantum well light-emitting layer **323** away from the N-type GaN layer **322**, the transparent conductive layer **325** is on a side of the P-type GaN layer **324** away from the quantum well light-emitting layer **323**, and the DBR layer **326** covers all surfaces of the light-emitting layer **32** except for a part of the surfaces contact with the base layer **33**.

[0022] The buffer layer **321** includes undoped GaN used to modulate crystal structure and reduce a lattice mismatch between the base layer **33** and the N-type GaN layer **322**. The N-type GaN layer **322** includes GaN doped with silicon (Si) and is used to provide electrons. The P-type GaN layer **324** is a high resistance material of GaN doped with magnesium (Mg), which is obtained after thermal annealing and is used to provide holes. The quantum well light-emitting layer **323** includes indium gallium nitride (InGaN) and GaN for emitting colored light, which can be blue, green, or red light. The transparent conductive layer **325** is used to form ohmic contact with the P-type GaN layer **324**, which is beneficial for input and output of current. The DBR layer **326** is composed of a series of alternating stacked high and low refractive index materials. The DBR layer **326** can achieve high reflection of light with a certain wavelength and low reflection of light with other wavelengths by periodic refractive index changes.

[0023] In this embodiment, the DBR layer **326** of the first sub-pixel **301** is used to reflect the first light, the second light, and the third light. The DBR layer **326** of the second sub-pixel **302** is used to transmit the first light and reflect the second light and the third light. The DBR layer **326** of the third sub-pixel **303** is used to transmit the first light and the second light and reflect the third light.

[0024] In other embodiments, each sub-pixel **30** may also be an organic light-emitting diode.

[0025] The conductive layer **31** includes a first contact electrode **311** and a second contact electrode **312** on a same side of the light-emitting layer **32**. The first contact electrode **311** and the second contact electrode **312** are spaced apart and on different edge parts of the light-emitting layer **32**. In this embodiment, the first contact electrode **311** is on a first edge **32a** of the light-emitting layer **32**, and the second contact electrode **312** is on a second edge **32b** of the light-emitting layer **32**. The first edge **32a** and second edge **32b** of the light-emitting layer **32** are both etched with openings, which connects the first contact electrode **311** to the transparent conductive layer **325**. The second contact electrode **312** is connected to the N-type GaN layer **322** for transmitting electrical signals to the light-emitting layer **32**. The first edge **32a** and the second edge **32b** of the light-emitting layer **32** are both etched with openings, which connects the first contact electrode **311** to the transparent conductive layer **325**. The second contact electrode **312** is connected to the N-type GaN layer **322** for transmitting electrical signals to the light-emitting layer **32**. The transparent conductive layer **325** transmits the electrical signals to the P-type GaN layer **324**, forming a voltage difference between the P-type GaN layer **324** and the N-type GaN layer **322**, thereby driving the light-emitting layer **32** to emit light. The first contact electrode **311** and the second contact electrode **312** can be metal electrodes.

[0026] The base layer **33** is transparent and non-conductive. A material of the base layer **33** can be sapphire, silicon carbide (SiC), or Si. The base layer **33** made of sapphire can reduce a cost and increase a size of the base layer **33**, so that an epitaxial layer can be heterogeneously grown on the substrate layer **33** as the light-emitting layer **32**. In other embodiments, the base layer **33** may be transparent plastic if each sub-pixel **30** is an organic light-emitting diode.

[0027] Referring to FIG. 2, each pixel **20** includes a plurality of metal circuit layers **34**. At least one metal circuit layer **34** is provided between the base layer **33** of the first sub-pixel **301** and the first contact electrode **311** and the second contact electrode **312** of the second sub-pixel **302**, and at least one metal circuit layer **34** is between the base layer **33** of the second sub-pixel **302** and the first contact electrode **311** and the second contact electrode **312** of the third sub-pixel **303**. Each metal circuit layer **34** is on a side of the base layer **33** away from the substrate **11**. Each metal circuit layer **34** connects the conductive layer **31** of one adjacent sub-pixel **30**. The base layer **33** of each sub-pixel **30** (except for the third sub-pixel **303**) contacts with at least one metal circuit layer **34**. The

base layers **33** of the first sub-pixels and the second sub-pixels are formed with a plurality of through holes **331** in an area which is not in contact with the light-emitting layers **32**. Pulse conducting electrodes **35** are formed on a side of each base layer **33** away from the metal circuit layer **34**, and the pulse conducting electrodes **35** extend through the through holes **331** to connect the metal circuit layers **34**.

[0028] In this embodiment, each pixel **20** includes three metal circuit layers **34**, namely a first metal circuit layer **341**, a second metal circuit layer **342**, and a third metal circuit layer **343**. The first metal circuit layer **341** is on a side of the base layer **33** of the first sub-pixel **301** away from the light-emitting layer **32**. The first metal circuit layer **341** is connected to the first contact electrode **311** and the second contact electrode **312** of the second sub-pixel **302**. The second metal circuit layer **342** is on a side of the base layer **33** of the second sub-pixel **302** away from the light-emitting layer **32**. The second metal circuit layer **342** is connected to the first contact electrode **311** and the second contact electrode **312** of the third sub-pixel **303**. The third metal circuit layer **343** is on a side of the base layer **33** of the first sub-pixel **301** away from the light-emitting layer **32**. The third metal circuit layer **343** is connected to the second sub-pixel **302**.

[0029] The pulse conducting electrodes **35** includes a first pulse conducting electrode **351**, a second pulse conducting electrode **352**, and a third pulse conducting electrode **353**. The first pulse conducting electrode **351** is on a side of the base layer **33** of the first sub-pixel **301** away from the first metal circuit layer **341**. The first pulse conducting electrode **351** is connected to the first metal circuit layer **341** through one through hole **331**. The second pulse conduction electrode **352** is on a side of the base layer **33** of the first sub-pixel **301** away from the third metal circuit layer **343**. The second pulse conduction electrode **352** is connected to the third pulse conducting electrode **353** through one through hole **331**. The third pulse conducting electrode **353** is between the base layer **33** and the third metal circuit layer **343** of the second sub-pixel **302**. The third pulse conducting electrode **353** is connected to the third metal circuit layer **343** and the second metal circuit layer **342** by extending through the through hole **331**.

[0030] The driving substrate **11** of the substrate **10** is on a side of the conductive layer **31** of the first sub-pixel **301** away from the light-emitting layer **32**. The driving substrate **11** is connected to the first contact electrode **311** and the second contact electrode **312** of the first sub-pixel **301**. The driving substrate **11** is used to transmit electrical signals to the first sub-pixel **301** to emit the first light. The driving substrate **11** connects the first contact electrode **311** and the second contact electrode **312** of the second sub-pixel **302** through the first pulse conducting electrode **351** and the first metal circuit layer **341**. The driving substrate **11** is used to transmit the electrical signals to the second sub-pixel **302** to emit the second light. The driving substrate **11** connects the first contact electrode **311** and the second contact electrode **312** of the third sub-pixel **303** through the second pulse conducting electrode **352**, the third metal circuit layer **343**, the third pulse conducting electrode **353**, and the second metal circuit layer **342**. The driving substrate **11** is used to transmit the electrical signals to the third sub-pixel **303** to emit the third light.

[0031] In this embodiment, the first sub-pixel **301**, the second sub-pixel **302**, and the third sub-pixel **303** have a same size on the plane vertical to the stacking direction. Along the stacking direction, the first light propagates upwards (towards human eyes) and passes through the base layer **33** of the first sub-pixel **301**, the second sub-pixel **302**, and the third sub-pixel **303**. The second light propagates upwards (towards human eyes) and passes through the base layer **33** of the second sub-pixel **302** and the third sub-pixel **303**. The third light propagates upwards (towards human eyes) and passes through the base layer **33** of the third sub-pixel **303**. The first light, the second light, and the third light are in different colors and are received by human eyes to show images to users.

[0032] In this embodiment, the first light is blue light, the second light is green light, and the third light is red light. In other embodiments, the first light, the second light, and the third light can be interchanged. For example, in other embodiments, the first light is red light, the second light is

green light, and the third light is blue light, or the first light is green light, the second light is blue light, and the third light is red light.

[0033] The driving substrate **11** includes a driving circuit **110** (see FIG. 4). The driving circuit **110** is used to control emission periods of the sub-pixels **30** through pulse modulation, so that different sub-pixels **30** in a same pixel **20** receive the electrical signals at different periods to emit light at different periods. The driving substrate **11** can include one driving circuit **110** to control the sub-pixels **30** simultaneously (see FIG. 4), or the driving substrate **11** can include a plurality of driving circuits **110** to control the sub-pixels **30**, respectively (see FIG. 5).

[0034] A driving pulse period T of the driving circuit **110** includes three periods, namely a first period T1, a second period T2, and a third period T3. The driving circuit **110** drives the first sub-pixel **301** to emit the first light during the first period T1, drives the second sub-pixel **302** to emit the second light during the second period T2, and drives the third sub-pixel **303** to emit the third light during the third period T3. That is, during the driving pulse period T, the driving circuit **110** sequentially drives the first sub-pixel **301**, the second sub-pixel **302**, and the third sub-pixel **303** to emit light, causing the three sub-pixels **30** in a same pixel **20** to emit different colors of light during different periods. Based on a visual persistence principle, the human eyes can receive image light mixed with three colors of light (the first light, the second light, and the third light) if a frequency of the driving pulse is greater than 120 Hz.

[0035] Each pixel **20** of the display panel **100** in this embodiment is composed of three sub-pixels **30**. The first sub-pixel **301**, the second sub-pixel **302**, and the third sub-pixel **303** are sequentially connected through coaxial stacking, so that the first sub-pixel **301**, the second sub-pixel **302**, and the third sub-pixel **303** are on different planes, which is conducive to reducing a volume of each pixel **20**, facilitating the display panel **100** applying on small devices such as AR glasses, and enhancing user experience.

Second Embodiment

[0036] Referring to FIG. 3, different from the first embodiment, each pixel **20** of the display panel **100** in this embodiment includes two sub-pixels **30** (i.e., the first sub-pixel **301** and the second sub-pixel **302**). The first sub-pixel **301** and the second sub-pixel **302** are coaxially stacked along the thickness direction. The first sub-pixel **301**, the second sub-pixel **302**, and the driving substrate **11** connect as the same with the first embodiment, and the driving substrate **11** drives the first sub-pixel **301** and the second sub-pixel **302** in the same way as described in the first embodiment.

[0037] Along the stacking direction, the first light emitted by the first sub-pixel **301** propagates upwards (towards the human eye) and passes through the base layer **33** of the first sub-pixel **301** and the second sub-pixel **302**. The second light emitted by the second sub-pixel **302** propagates upwards (towards the human eye) and passes through the base layer **33** of the second sub-pixel **302**. The first light and the second light are in different colors and are received by the human eyes to show images to the users. In this embodiment, the first light is green light, and the second light is red light. In other embodiments, the first light may be red light, and the second light may be green light.

[0038] Each pixel **20** in this embodiment includes two sub-pixels **30** to display images, which

[0039] has low energy consumption, longer service life, as well as high luminous efficiency, and can achieve high brightness display effects.

[0040] In summary, the embodiments of this application have the following beneficial effects:

[0041] In the display panels **100** above, all sub-pixels **30** in a same pixel **20** are coaxially stacked along the stacking direction (i.e. the thickness direction). In each pixel **20**, the sub-pixel **30** at the bottom (i.e. the first sub-pixel **301** in the first embodiment and the second embodiment) connects the driving substrate **11**, the remaining sub-pixels **30** connect the driving substrate **11** through the pulse conducting electrodes **35** and the metal circuit layers **34**. Sub-pixels **30** in a same pixel **20** connect the driving substrate **11** respectively, so that the driving substrate **11** can transmit electrical signals to the sub-pixels **30** respectively. The driving circuit **110** of the driving substrate

11 controls the emission periods of the sub-pixels **30** in a same pixel **20** through pulse modulation, so that the sub-pixels **30** in a same pixel **20** receives the electrical signals and emit lights during different periods. Based on the visual persistence principle, the human eye can recognize the image light mixed with multiple colors of light when the pulse frequency of the pulse modulation is greater than a recognition frequency of the human eye. Compared to display panels in the art, the sub-pixels **30** in each pixel **20** of the display panel **100** of the present disclosure are not on a same plane and are stacked coaxially, which is conducive to reducing the volume of each pixels **20** of the display panel **100**, simplifying a production process of the display panels **100**, and increasing the producing yield.

[0042] Finally, it should be noted that the above embodiments are only used to illustrate the technical solution of the present application and not to limit the present application. Although the present application has been described in detail with reference to preferred embodiments, one ordinary skill in the art should understand that the technical solution of the present application can be modified or equivalent replaced without departing from the spirit and scope of the technical solution of the present application.

Claims

1. A display panel comprising: a substrate; and a plurality of pixels on the substrate, each of the plurality of pixels comprising a plurality of sub-pixels coaxially stacked, each of the plurality of sub-pixels comprising a conductive layer, a light-emitting layer electrically connected to the conductive layer, and a base layer, the plurality of pixels arranging in an array on a plane vertical to a stacking direction of the plurality of sub-pixels; wherein the conductive layer is configured to transmit an electrical signal, the light-emitting layer is configured to emit light in response to the electrical signal, and the plurality of sub-pixels is stacked in a sequence, along the stacking direction, the base layers of the sub-pixels with a same sequence are connected with each other.
2. The display panel according to claim 1, wherein the conductive layer comprises a first conducting electrode and a second conducting electrode on a same side of the light-emitting layer, and the base layer contacts a side of the light-emitting layer away from the conductive layer; and each of the plurality of pixels comprises a plurality of metal circuit layers on a side of the base layer away from the light-emitting layer, the plurality of metal circuit layers connect the conductive layer of an adjacent pixel, a part of the base layer contact at least one of the plurality of metal circuit layers and not contact the light-emitting layer is formed with a plurality of through holes, at least one pulse conducting electrode is on a side of the base layer away from the plurality of metal circuit layers, and the at least one pulse conducting electrode connects to at least one of the plurality of metal circuit layers through at least one of the plurality of through holes.
3. The display panel according to claim 2, wherein the substrate comprises a driving substrate; the driving substrate is at a side of the conductive layer of the sub-pixel at the bottom away from the light-emitting layer; and the driving substrate connects the conductive layer and is configured to send electrical signals to the plurality of sub-pixels to drive the plurality of sub-pixels to emit light.
4. The display panel according to claim 3, wherein the driving substrate connects at least one of the plurality of pulse conducting electrodes, each of the plurality of pulse conducting electrode is configured to send the electrical signal to the plurality of sub-pixels in each of the plurality of pixels except for a bottommost sub-pixel.
5. The display panel according to claim 4, wherein the driving substrate is configured to control the plurality of sub-pixels to emit light during different periods by pulse modulation.
6. The display panel according to claim 5, wherein the driving substrate comprises one driving circuit to control the plurality of sub-pixels.
7. The display panel according to claim 5, wherein the driving substrate comprises a plurality of driving circuits to control the plurality of sub-pixels.

- 8.** The display panel according to claim 1, wherein the plurality of sub-pixels in each of the plurality of pixels emit light of different colors.
- 9.** The display panel according to claim 8, wherein the light-emitting layer is configured to emit red light, green light, or blue light according to the electrical signal.
- 10.** The display panel according to claim 1, wherein the light from the light-emitting layer transmits along the stacking direction away from the substrate.
- 11.** The display panel according to claim 1, wherein the base layer of the sub-pixel at the top covers the light-emitting layer of the sub-pixel at the top.
- 12.** The display panel according to claim 1, wherein each of the plurality of pixels comprises a first sub-pixel and a second sub-pixel coaxially stacked along the stacking direction; and the base layers of the first sub-pixels of the plurality of pixels are connected with each other, and the base layers of the second sub-pixels of the plurality of pixels are connected with each other.
- 13.** The display panel according to claim 12, wherein each of the plurality of pixels further comprises a third sub-pixel on a side of the second sub-pixel away from the first sub-pixel, the first sub-pixel, the second sub-pixel, and the third sub-pixel are coaxially stacked along the stacking direction; and the base layers of the third sub-pixels of the plurality of pixels are connected with each other.
- 14.** The display panel according to claim 12, wherein the first sub-pixel is configured to emit red light, and the second sub-pixel is configured to emit green light; or the first sub-pixel is configured to emit green light, and the second sub-pixel is configured to emit red light.
- 15.** The display panel according to claim 12, wherein the first sub-pixel is configured to emit a first light, and the second sub-pixel is configured to emit a second light; and the first light passes through the second sub-pixel to show an image.
- 16.** The display panel according to claim 12, wherein the first light is red light and the second light is green light; or The first light is green light and the second light is red light.
- 17.** The display panel according to claim 1, wherein the plurality of sub-pixels have a same size on the plane vertical to the stacking direction.
- 18.** A display panel comprising: a substrate; and a plurality of pixels on the substrate, each of the plurality of pixels comprising a plurality of sub-pixels coaxially stacked, each of the plurality of sub-pixels comprising a conductive layer, a base layer and a light-emitting layer connected between the conductive layer and the base layer, the plurality of pixels arranging in an array on a plane vertical to a stacking direction of the plurality of sub-pixels; wherein the conductive layer is configured to transmit an electrical signal, the light-emitting layer is configured to emit light according to the electrical signal, and the plurality of sub-pixels have a same size on the plane vertical to the stacking direction.
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